

Dehn surgeries and $SU(2)$

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Background: The Kirby Problem

Kirby Problem 3.105 (A)

- When is a 3-manifold Y $SU(2)$ non-abelian?
- Is any integer homology sphere ($\mathbb{Z}HS^3$) $SU(2)$ non-abelian?

Definition

A manifold Y is $SU(2)$ **non-abelian** if there exists a homomorphism

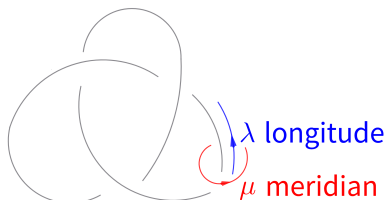
$$\rho : \pi_1(Y) \rightarrow SU(2)$$

such that $\text{Im}(\rho)$ is non-abelian.

Previous Results

Theorem (Kronheimer–Mrowka '04)

For any nontrivial knot $K \subset S^3$ and rational $r \in [0, 2]$, the (Dehn) surgery manifold $S_r^3(K)$ is $SU(2)$ non-abelian.

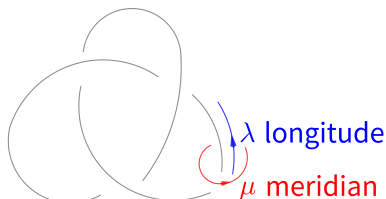


Right-handed trefoil $K = T_{2,3}$

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Theorem (Zentner '18)

The splicing manifold $Y(K_1, K_2) = (S^3 \setminus \nu(K_1)) \cup_{\partial} (S^3 \setminus \nu(K_2))$ (gluing meridian to longitude and vice versa) is $SU(2)$ non-abelian.



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- Both $S_{1/n}^3(K)$ and $Y(K_1, K_2)$ are $\mathbb{Z}HS^3$.
- We only consider $r \geq 0$ because for the mirror knot $m(K)$ of K ,

$$\pi_1(S_{-r}^3(K)) \cong \pi_1(S_r^3(m(K)))$$

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- Kronheimer–Mrowka's theorem implies $\pi_1(S_r^3(K))$ is non-abelian for $r \in [0, 2]$. The case $r = 1$ confirms the **“Property P” conjecture**.
- Zentner further showed (in the same paper) that any $\mathbb{Z}HS^3 Y$ is $SL(2, \mathbb{C})$ non-abelian (based on Geometrization Theorem).

Question on Larger Slopes

Question (Kronheimer–Mrowka '04)

Is $S_r^3(K)$ always $SU(2)$ non-abelian for $r = 3, 4$?

Remark

This is **NOT** always true for $r = 5$. For the right-handed trefoil $T_{2,3}$:

$$S_5^3(T_{2,3}) = L(5, 4)$$

Here $\pi_1 = \mathbb{Z}/5$, which is $SU(2)$ abelian.

Detailed Progress

Assuming p is **prime power**. A rational slope r for $S_r^3(K)$ is $SU(2)$ non-abelian in the following cases:

- $r = 4$ and $p/q \in (2, 3)$ (Baldwin–Sivek '23)
- $r = 3$ (Baldwin–Li–Sivek–Y. '24)
- $r = p/q \in (2, 5)$ (Farber–Reinoso–Wang '24)
- $r = 5$, when $K \neq T_{2,3}$ (Li–Y. '25a)
- $r = p/q$, $2p/q \in (2, 6)$, when $(K, r) \neq (T_{2,3}, 6 - 1/n)$ for $n \in \mathbb{Z}_+$ (Li–Y. '25b)
- **Depending on unpublished work:** $r = p/q$, $2p/q \in (2, 8]$ when $(K, r) \neq (T_{2,3}, 6 - 1/n)$ for $n \in \mathbb{Z}$ (Ghosh–Miller–Eismeier '25)
- **Conjecture (Y.):** Holds for $r = p/q$, $2p/q \in (2, 12]$ with exceptions $(T_{2,3}, 6 - 1/n)$ and $(T_{2,5}, 10 - 1/n)$ for $n \in \mathbb{Z}$

Tool: Framed Instanton Homology

The condition p **prime power** allows us to use framed instanton homology $I^\sharp(Y)$ (Kronheimer–Mrowka '11) defined for **ANY** closed 3-manifold Y .

- $I^\sharp(Y)$ is a $\mathbb{Z}/4$ -graded $\mathbb{Z}$ -module.
- Euler characteristic (Scaduto '15):

$$\chi(I^\sharp(Y)) = \begin{cases} |H_1(Y; \mathbb{Z})| & b_1(Y) = 0 \\ 0 & b_1(Y) > 0 \end{cases}$$

Remark

Floer's original homology $I(Y)$ was only defined for $\mathbb{Z}HS^3$ with $\chi(I(Y)) = \text{Casson}(Y)$.

- Roughly, $I(Y)$ is related to $H_*(X(Y))$,
- while $I^\sharp(Y)$ is related to $H_*(R(Y))$,

where $R(Y) = \text{Hom}(\pi_1(Y), SU(2))$ and $X(Y) = R(Y)/SU(2)$.

Proof Strategy: Spectral Sequence

Theorem (Baldwin–Sivek '23 for p/q , Li–Y. '25 for $2p/q$)

Let p be a prime power, $r = p/q$, $2p/q > 0$. If $Y = S_r^3(K)$ is **NOT** $SU(2)$ non-abelian, then there exists a spectral sequence:

$$H_*(R(Y)) \implies I^\sharp(Y)$$

Recall that $R(Y) = \text{Hom}(\pi_1(Y), SU(2))$.

Corollary

In such a case (only abelian representations $\rho : \pi_1(Y) \rightarrow SU(2)$),

$$I^\sharp(Y) \cong \mathbb{Z}^{|H_1(Y; \mathbb{Z})|}$$

We will sketch the proof of the corollary.

Proof of Corollary

Corollary (restated)

Let p be a prime power, $r = p/q$, $2p/q > 0$. If $Y = S_r^3(K)$ is **NOT** $SU(2)$ non-abelian, then

$$I^\sharp(Y) \cong \mathbb{Z}^{|H_1(Y; \mathbb{Z})|}$$

- For $r = p/q$, $H_1(Y; \mathbb{Z}) = \mathbb{Z}/p$.
- If all $\rho : \pi_1(Y) \rightarrow SU(2)$ have abelian images, then

$$\begin{aligned} X(Y) &= R(Y)/SU(2) \\ &= \text{Hom}(\mathbb{Z}/p, SU(2))/SU(2) \\ &= \{[\rho_k]\}_{k=0, \dots, [p/2]}, \end{aligned}$$

where ρ_k is the abelian representation sending $1 \in \mathbb{Z}/p$ to

$$\begin{bmatrix} e^{2\pi i k/p} & 0 \\ 0 & e^{-2\pi i k/p} \end{bmatrix}$$

Proof of Corollary

Corollary (restated)

Let p be a prime power, $r = p/q$, $2p/q > 0$. If $Y = S_r^3(K)$ is **NOT** $SU(2)$ non-abelian, then

$$I^\sharp(Y) \cong \mathbb{Z}^{|H_1(Y; \mathbb{Z})|}$$

- The stabilizer of ρ_k is:

$$\text{Stab}(\rho_k) = \begin{cases} SU(2)/\pm \text{Id} & k = 0, p/2 \text{ (if } p \text{ even)} \\ U(1) & \text{otherwise} \end{cases}$$

- The pre-images of $[\rho_k]$ in $R(Y)$ are $(SU(2)/\pm \text{Id})/\text{Stab}(\rho_k)$. Thus:

$$R(Y) = \text{Hom}(\mathbb{Z}/p, SU(2)) = \begin{cases} 1 \text{ pt} \cup \frac{p-1}{2} S^2 & p \text{ odd} \\ 2 \text{ pts} \cup \frac{p-2}{2} S^2 & p \text{ even} \end{cases}$$

- We compute $H_*(R(Y)) \cong \mathbb{Z}^p$.

Proof of Corollary

Corollary (restated)

Let p be a prime power, $r = p/q$, $2p/q > 0$. If $Y = S_r^3(K)$ is **NOT** $SU(2)$ non-abelian, then

$$I^\sharp(Y) \cong \mathbb{Z}^{|H_1(Y; \mathbb{Z})|}$$

- We compute $H_*(R(Y)) \cong \mathbb{Z}^p$.
- Since $\chi(I^\sharp(Y)) = |H_1(Y; \mathbb{Z})| = p$ and the rank matches, the spectral sequence $H_*(R(Y)) \implies I^\sharp(Y)$ must **collapse**.
- Since $H_*(R(Y))$ is free, the S.S. over other coefficients also collapses. Thus, we obtain $I^\sharp(Y) \cong \mathbb{Z}^p$.

Instanton L-space

Definition

A closed 3-manifold Y with $b_1(Y) = 0$ is called an **instanton L-space** if:

$$\text{rank } I^\#(Y) = |H_1(Y; \mathbb{Z})|$$

Definition

A knot K is called an **instanton L-space knot** if for some $r > 0$, $S_r^3(K)$ is an instanton L-space.

Remark

The notion “L-space” comes from Heegaard Floer theory (Ozsváth–Szabó) as a generalization of lens spaces.

Instanton L-space and Slope Constraints

Corollary

Let p be a prime power, $r = p/q$, $2p/q > 0$. If $Y = S_r^3(K)$ is **NOT** $SU(2)$ non-abelian, then:

- Y is an instanton L-space.
- K is an instanton L-space knot.

Theorem (Baldwin–Sivek '24)

If K is an instanton L-space knot, then $S_r^3(K)$ is an instanton L-space if and only if:

$$r \geq 2g(K) - 1,$$

where $g(K)$ is the Seifert genus of K .

Proof Idea for Main Theorem

Goal: Prove $SU(2)$ non-abelian for $r \in (2, 5)$ (consider $r \in [5, 6)$ later).

Contradiction: If **NOT** non-abelian $\implies r \geq 2g(K) - 1$.

- If $g(K) \geq 3$, then $2g - 1 \geq 5$. So for $r < 5$, we have a contradiction immediately.
- We only need to check instanton L-space knots of small genus.

Small Genus Cases:

- $g = 1$: Only $T_{2,3}$ (Baldwin–Sivek).
- $g = 2$: Only $T_{2,5}$ (Baldwin–Li–Sivek–Y., Farber–Reinoso–Wang).

Exceptions: These specific cases (for $T_{2,3}$ and $T_{2,5}$) are checked to be $SU(2)$ non-abelian (Sivek–Zentner '22).

Beyond Slope 5: $\mathbb{Z}/2$ Coefficients

- **Idea for slopes beyond 5:** Consider $I^\sharp(Y; \mathbb{Z}/2)$. Note that

$$\text{rank } I^\sharp(Y) = \dim I^\sharp(Y; \mathbb{C}).$$

- **Conceptual reason:** $R(Y) = \text{Hom}(\pi_1(Y), SU(2))$ can contain copies of $SU(2)/\pm \text{Id} = SO(3)$. 2-torsion appears in

$$H_*(SO(3); \mathbb{Z}) \cong \mathbb{Z} \oplus \mathbb{Z} \oplus \mathbb{Z}/2.$$

Corollary (similar proof as before)

Let p be a prime power, $r = p/q$, $2p/q > 0$. If $Y = S_r^3(K)$ is **NOT** $SU(2)$ non-abelian, then:

- Y is an instanton L-space over $\mathbb{Z}/2$ ($\dim I^\sharp(Y; \mathbb{Z}/2) = p$).
- K is an instanton L-space knot over $\mathbb{Z}/2$ (similar definition).

Beyond Slope 5: $\mathbb{Z}/2$ Coefficients

Theorem (Li–Y. '25)

If K is an instanton L-space knot over $\mathbb{Z}/2$, then $S_r^3(K)$ is an instanton L-space over $\mathbb{Z}/2$ for any

$$r > \nu_2^\sharp(K),$$

where $\nu_2^\sharp(K) = \nu_{\mathbb{Z}/2}^\sharp(K)$ is an integer invariant for any $K \subset S^3$ and $\nu_2^\sharp(K) \geq 2g(K)$ when K is an instanton L-space knot over $\mathbb{Z}/2$.

- The proof for $r \in [5, 6)$ relies on this sharper bound.
- $\nu_{\mathbb{Z}/2}^\sharp(K)$ has generalization over any field $\mathbb{F}$ (Li–Y. '25).
 $\nu_{\mathbb{F}}^\sharp(K)$ is a **concordance invariant** with the property for mirror knot:

$$\nu_{\mathbb{F}}^\sharp(m(K)) = -\nu_{\mathbb{F}}^\sharp(K)$$

- $\nu_{\mathbb{C}}^\sharp(K)$ is previously studied by Baldwin–Sivek '21 and motivates $\nu_{\mathbb{F}}^\sharp(K)$. $\nu_{\mathbb{C}}^\sharp(K) = 2g(K) - 1$ for instanton L-space knot K .

Thanks for your attention.
Slides on floer.wiki/en